



HTML5

Session1

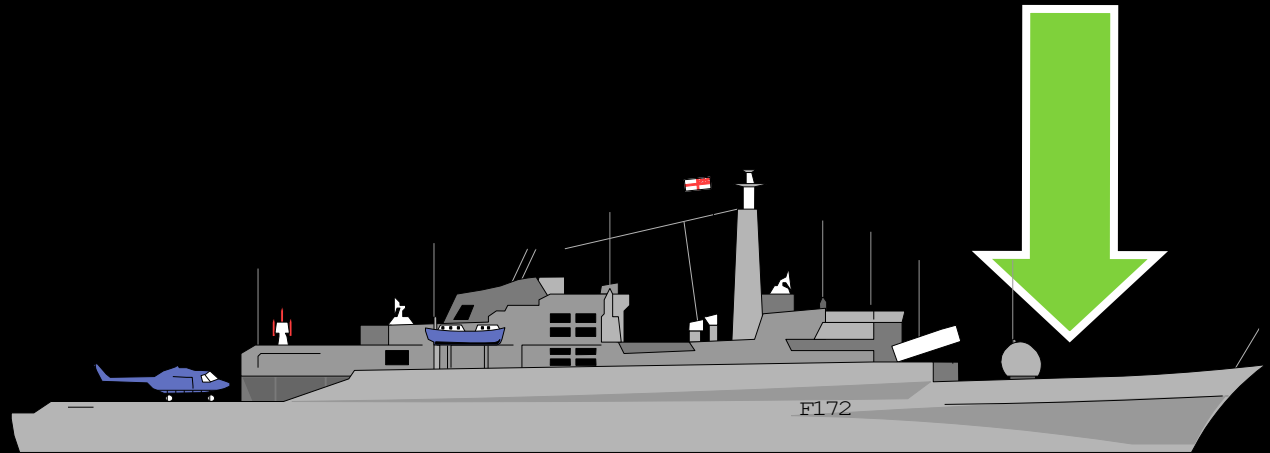
Microsoft
Be what's next.™

Microsoft® Student Partners



Direct Marketing for Business

Ford in US Market



Ford to Indian Market and
then it may get sold



Advantages of Net Marketing

- Lower cost of product advertisement
 - Removal of time restriction from business deals
 - Larger audience reach
- 



Internet

- It means connecting a computer to any other computer anywhere in the world via dedicated routers and servers.
 - When two computers are connected over the Internet, they can send and receive all kinds of information such as text, graphics, voice, video, and computer programs.
 - No one owns Internet, although several organizations the world over collaborate in its functioning and development.
 - The high-speed, fiber-optic cables (called backbones) through which the bulk of the Internet data travels are owned by telephone companies in their respective countries.
- 



World Wide Web (WWW)

- The World Wide Web (WWW) is combination of all resources and users on the Internet that are using the Hypertext Transfer Protocol (HTTP).
 - A broader definition comes from the World Wide Web Consortium (W3C):
 - "The World Wide Web is the universe of network-accessible information, an embodiment of human knowledge."
- 



Web Protocols

- The Internet relies on a number of protocols in order to function properly.
 - A protocol is simply a standard for enabling the connection, communication, and data transfer between two places on a network.
 - HTTP
 - HTTPS
 - FTP
 - SMTP
- 



HTTP

- HTTP stands for Hypertext Transfer Protocol. It is the standard protocol for transferring web pages (and their content) across the Internet.
 - When you browse a web page, the URL might be preceded by `http://`.
 - This is telling the web browser to use HTTP to transfer the data.
 - Browsers will default to HTTP if you don't specify it. You can test this by typing in say...
`www.quackit.com` (instead of `http://www.quackit.com`).
- 



HTTPS

- HTTPS stands for Hypertext Transfer Protocol over Secure Socket Layer.
 - It is a secure version of HTTP.
 - HTTPS is used primarily on web pages that ask you to provide personal or sensitive information (such as a password or your credit card details).
 - When you browse a web page using HTTPS, you are using SSL (Secure Sockets Layer).
- 



FTP

- FTP stands for File Transfer Protocol.
 - It is used to transfer files across the Internet.
 - FTP is commonly used by web developers to publish updates to a website (i.e. to upload a new version of the website).
 - Where HTTP is used for displaying the file in your browser, FTP is used simply to transfer the file from one computer to a specified location on another computer.
 - use FTP to transfer the files from your computer to a remote computer (such as a web server), or to transfer from the remote computer to your local computer.
- 

Simple Mail Transfer Protocol (SMTP)

- SMTP is an application layer protocol.
- The client who wants to send the mail opens a TCP connection to the SMTP server and then sends the mail across the connection.
- The SMTP server is always on listening mode.
- As soon as it listens for a TCP connection from any client, the SMTP process initiates a connection on that port (25).
- After successfully establishing the TCP connection the client process sends the mail instantly.



Web Standards

- Web standards are rules and guidelines established by the World Wide Web Consortium (W3C) developed to promote consistency in the design code which makes up a web page.
 - Without getting technical, simply it's the guideline for the mark-up language which determines how a web page displays in a visitor's browser window.
- 



Why Web Standards ?

- Web pages will display in a wide variety of browsers and computers, including new technology like iPhones, Droids, iPads, PDA devices, mobile phones, which greatly increases the viewing audience.
 - W3C Standards promote the use of "Cascading Style Sheets" (CSS) or design code which is attached to the web page. rather than embedded in the page.
 - CSS significantly reduces the page file size which means not only a faster page loading time but lower hosting costs for frequently visited sites due to using less bandwidth.
 - Search Engines are able to access and index pages designed to web standards with greater efficiency.
- 

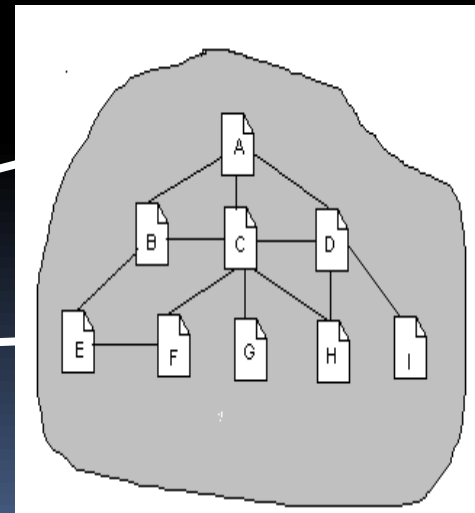


Common Web Standards

- Recommendations for markup languages, such as Hypertext Markup Language (HTML), Extensible Hypertext Markup Language (XHTML), and Scalable Vector Graphics (SVG) from W3C.
 - Recommendations for stylesheets, especially Cascading Style Sheets (CSS), from W3C.
 - Standards for ECMAScript, more commonly JavaScript, from Ecma International.
 - Recommendations for Document Object Models (DOM), from W3C.
 - Properly formed names and addresses for the page and all other resources referenced from it (URIs), based upon RFC 2396, from IETF.[12]
 - Proper use of HTTP and MIME to deliver the page, return data from it and to request other resources referenced in it, based on RFC 2616, from IETF.[13]
- 

Web Site and Web Page

- The first page which opens in a Web Site is the HOME page
- There is no restriction with respect to the site size
- One web site can span over more than one server also





Features of Web Pages

Web pages allow the following features :

- Product Catalogues
 - Web Chat
 - Online order tracking
 - Online Demonstration
 - Online Feedback System
- 



General Guidelines

- Avoid use of too many fonts
- Avoid too many colours on the same page
- Use pastoral colours in corporate and professional slides
- Use bright and sharp hues in kids and entertainment slides



Phases of developing a Web site

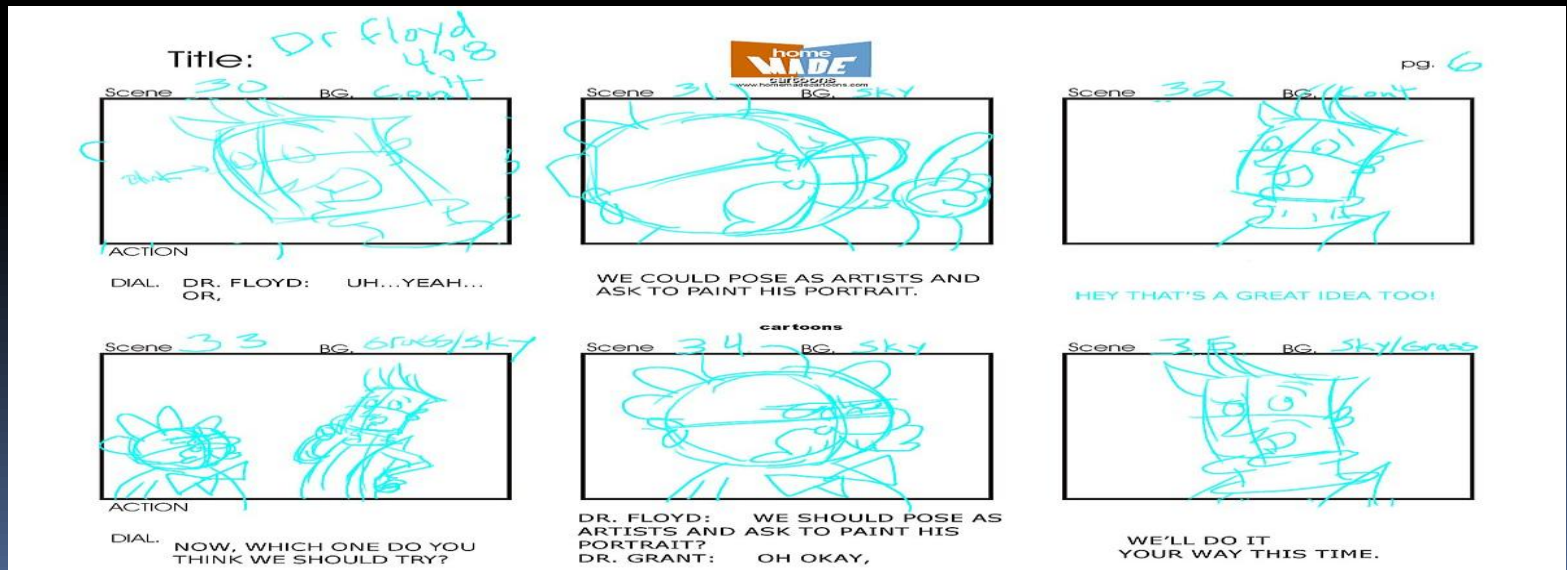
The web site can span various phases :

- Design Phase
 - Development Phase
- 

The Design Phase

The design phase covers :

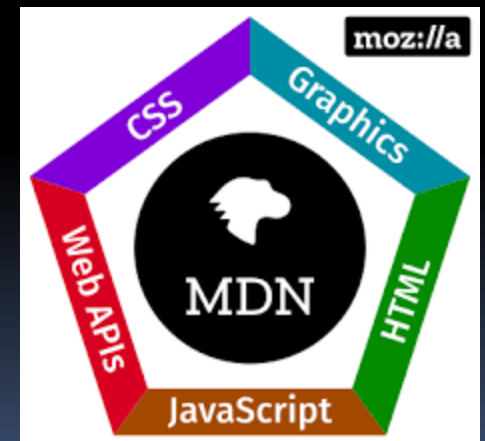
- The Goals and Objectives
- Target Audience
- Storyboarding
- Organize your information



Development Phase

The development phase covers :

- The layout of the web page and the web site
 - Unity and Variety
 - Recognizing the Medium
 - Effect and Layout
 - Using color
 - Background
 - Text
- Navigation around the web site
 - Hyper tree
 - Image maps
 - Hyperlinks
 - Sub-pages
 - Table of Contents



What is HTML?

- HyperText Markup Language is the language for specifying the *static* content of Web pages.
- HTML is a language for describing web pages.
- HTML stands for
HyperText Markup Language

HTML Introduction

- HTML is a **markup** language
- A markup language is a set of markup **tags**
- The tags **describe** document content
- HTML documents contain HTML **tags** and plain **text**
- HTML documents are also called **web pages**



HTML Tags

- HTML tags are keywords (tag names) surrounded by **angle brackets** like <html>
- HTML tags normally **come in pairs** like <p> and </p>
- The first tag in a pair is the **start tag**, the second tag is the **end tag**
- The end tag is written like the start tag, with a **slash** before the tag name
- Start and end tags are also called **opening tags** and **closing tags**

<tagname>content</tagname>

HTML Elements

- In HTML, most elements are written with a start tag (e.g. <p>) and an end tag (e.g. </p>), with the content in between:

```
<title> HTML Document</title>  
<H1>This is Heading 1</H1>  
<p>This is a paragraph.</p>
```



Web Browsers

- It uses the tags to determine how the content of the HTML page is to be presented/displayed to the user.
 - The browser does not display the HTML tags.
 - The purpose of a web is to read HTML documents and display them as web pages.
 - **Browser is a software, used to display Web Pages.**
 - Eg for Browsers: Google Chrome, Internet Explorer, Firefox, Safari.
- 

HTML Page Structure

```
<html>
```

```
<head>
```

```
</head>
```

```
<body>
```

```
<h1>This a heading</h1>
```

```
<p>This is a paragraph.</p>
```

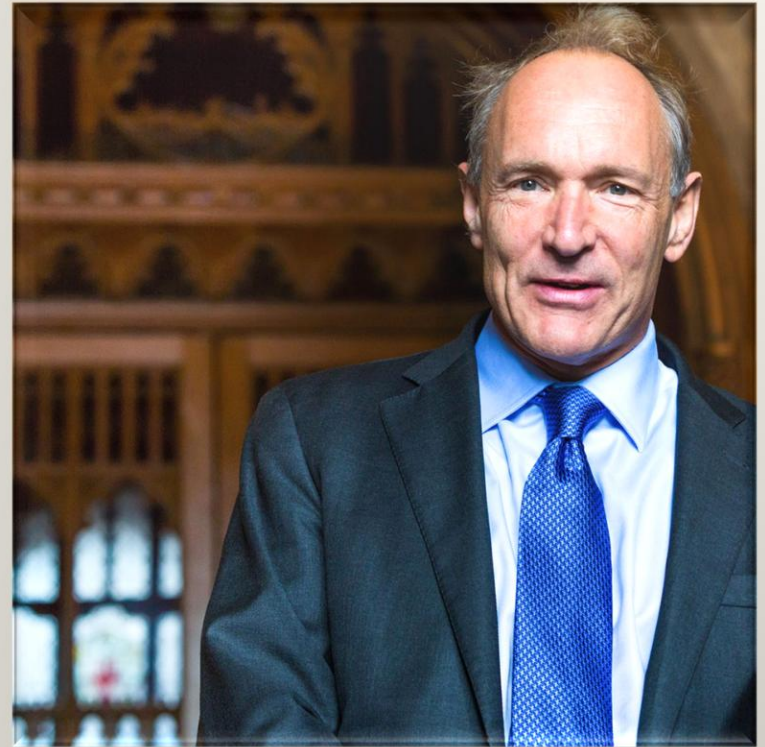
```
<p>This is another paragraph.</p>
```

```
</body>
```

```
</html>
```

HISTORY OF HTML

- Tim Berners-Lee, a physicist at the CERN research institute in Switzerland invented HTML in 1991.
- This first version consisted of 18 HTML tags.



HTML Versions

Version	Year
HTML	1991
HTML+	1993
HTML 2.01	Nov 1995
HTML 3.2	Jan 1997
HTML 4.01 – Widely Used & Successful	Dec 1999
XHTML 1.0	May 2000
HTML5	2012
XHTML	2012
HTML5 W3C RECOMMENDATION	Oct 2014
HTML 5.1 , 5.2 W3C	Nov 2016 , Dec 2017



HTML Editors

HTML can be edited by using a professional HTML editor like:

- Adobe Dreamweaver
 - Microsoft Expression Web
 - Coffee Cup HTML Editor
 - Ms Visual Studio Code Editor
 - However, for learning HTML we recommend a text editor like Notepad (PC) or TextEdit (Mac).
- 

What is HTML5?



- HTML5 is the latest standard for HTML.
- The previous version of HTML, HTML 4.01, came in 1999, and the internet has changed significantly since then.
- HTML5 was designed to replace both HTML 4, XHTML, and the HTML DOM Level 2.
- It was specially designed to deliver rich content without the need for additional plugins.
- The current version delivers everything from animation to graphics, music to movies, and can also be used to build complicated web applications.
- HTML5 is also cross-platform. It is designed to work whether you are using a PC, or a Tablet, a Smartphone, or a Smart TV.



How Did HTML5 Get Started?

- HTML5 is a cooperation between the World Wide Web Consortium (W3C) and the Web Hypertext Application Technology Working Group (WHATWG).
 - WHATWG was working with web forms and applications, and W3C was working with XHTML 2.0.
 - In 2006, they decided to cooperate and create a new version of HTML.
- 



Rules for HTML5

- New features should be based on HTML, CSS, DOM, and JavaScript
 - The need for external plugins (like Flash) should be reduced
 - Error handling should be easier than in previous versions
 - Scripting has to be replaced by more markup
 - HTML5 should be device-independent
 - The development process should be visible to the public
- 

The `<!DOCTYPE>` Declaration

- The `<!DOCTYPE>` declaration helps the browser to display a web page correctly.
- There are many different documents on the web, and a browser can only display an HTML page 100% correctly if it knows the HTML version and type used.
- In HTML5 there is only one DOCTYPE declaration, and it is very simple:

```
<!DOCTYPE html>
```


A Minimum HTML5 Document

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Title of the document</title>
</head>

<body>
Content of the document.....
</body>

</html>
```

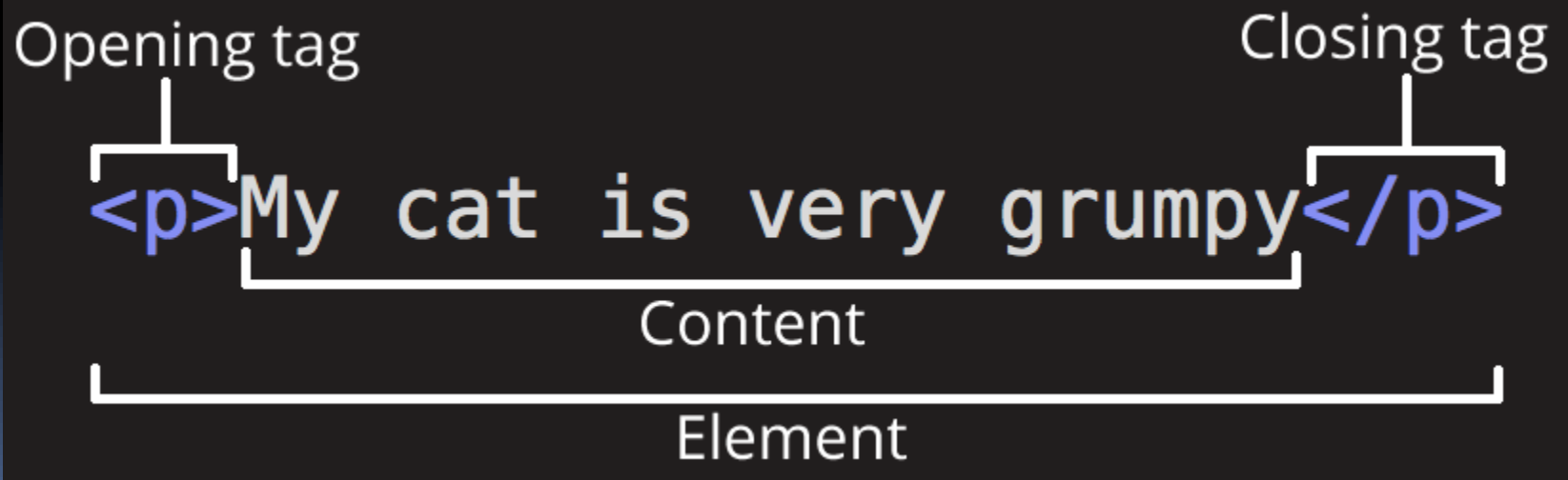


HTML5 - New Features

- The <canvas> element for 2D drawing
 - The <video> and <audio> elements for media playback
 - Support for local storage
 - New content-specific elements, like <article>, <footer>, <header>, <nav>, <section>
 - New form controls, like calendar, date, time, email, url, search
- 

HTML Elements

- HTML documents are defined by HTML elements.
- An HTML element is everything from the start tag to the end tag.





HTML Element Syntax

- An HTML element starts with a **start tag / opening tag**
 - An HTML element ends with an **end tag / closing tag**
 - The **element content** is everything between the start and the end tag
 - Some HTML elements have **empty content**
 - Empty elements are **closed in the start tag**
 - Most HTML elements can have **attributes**
- 

Nested HTML Elements

- Most HTML elements can be nested (can contain other HTML elements).
- HTML documents consist of nested HTML elements.

The <p> element
The <body> element
The <html> element

Empty HTML Elements

- HTML elements with no content are called empty elements.
- `<br>` is an empty element without a closing tag (the `<br>` tag defines a line break).
- In XHTML, all elements must be closed.
- Adding a slash inside the start tag, like `<br />`, is the proper way of closing empty elements in XHTML (and XML).



HTML Attributes

- Attributes provide additional information about HTML elements.
 - HTML elements can have **attributes**
 - Attributes provide **additional information** about an element
 - Attributes are always specified in **the start tag**
 - Attributes come in name/value pairs like: **name="value"**
- 

Attribute Example

- Attribute values should always be enclosed in quotes.
- HTML links are defined with the `<a>` tag.
- The link address is specified in the **href attribute**:

```
<a href="http://www.newhorizons.com">This is a link</a>
```


HTML Headings

- Headings are important in HTML documents.
- Headings are defined with the `<h1>` to `<h6>` tags.
- `<h1>` defines the most important heading.
`<h6>` defines the least important heading.

`<h1>This is a heading</h1>`

`<h2>This is a heading</h2>`

`<h3>This is a heading</h3>`



Headings...

- Use HTML headings for headings only.
 - Don't use headings to make text **BIG** or **bold**.
 - Search engines use your headings to index the structure and content of your web pages.
 - Since users may skim your pages by its headings, it is important to use headings to show the document structure.
 - H₁ headings should be used as main headings, followed by H₂ headings, then the less important H₃ headings, and so on.
- 

HTML Lines

- The `<hr>` tag creates a horizontal line in an HTML page.
- The `hr` element can be used to separate contents.

```
<p>The hr tag defines a horizontal rule:</p>  
<hr>  
<p>This is a paragraph.</p>
```

HTML Paragraphs

- HTML documents are divided into paragraphs.
- Paragraphs are defined with the <p> tag.

<p>Happiness is not something ready made. It comes from your own actions.</p>

<p>There is only one happiness in this life, to love and be loved.</p>

HTML Formatting Tags

- HTML uses tags like `<b>` and `<i>` for formatting output, like **bold** or *italic* text.
- These HTML tags are called formatting tags

HTML Text Formatting

This text is bold

This text is italic

This is computer output

This is _{subscript} and ^{superscript}



`<strong>` & `<em>`

- Often `<strong>` renders as `<b>`
- `<em>` renders as `<i>`
- `<b>` or `<i>` defines bold or italic text only.
- `<strong>` or `<em>` means that you want the text to be rendered in a way that the user understands as "important".

HTML Text Formatting Tags

Tag	Description
<u></u>	Defines bold text
<u></u>	Defines emphasized text
<u><i></u>	Defines a part of text in an alternate voice or mood
<u><small></u>	Defines smaller text
<u></u>	Defines important text
<u><sub></u>	Defines subscripted text
<u><sup></u>	Defines superscripted text
<u><ins></u>	Defines inserted text
<u></u>	Defines deleted text
<u><mark></u>	Defines marked/highlighted text

Text Formatting Tags

Super Script - `<SUP> .... </SUP>`

`<P> This Is My 7 <SUP> th </SUP> Program Using HTML </P>`

Sub Script - `<SUB> .... </SUB>`

`<P> H<SUB>2</SUB>O Is The Chemical Name For Water</P>`

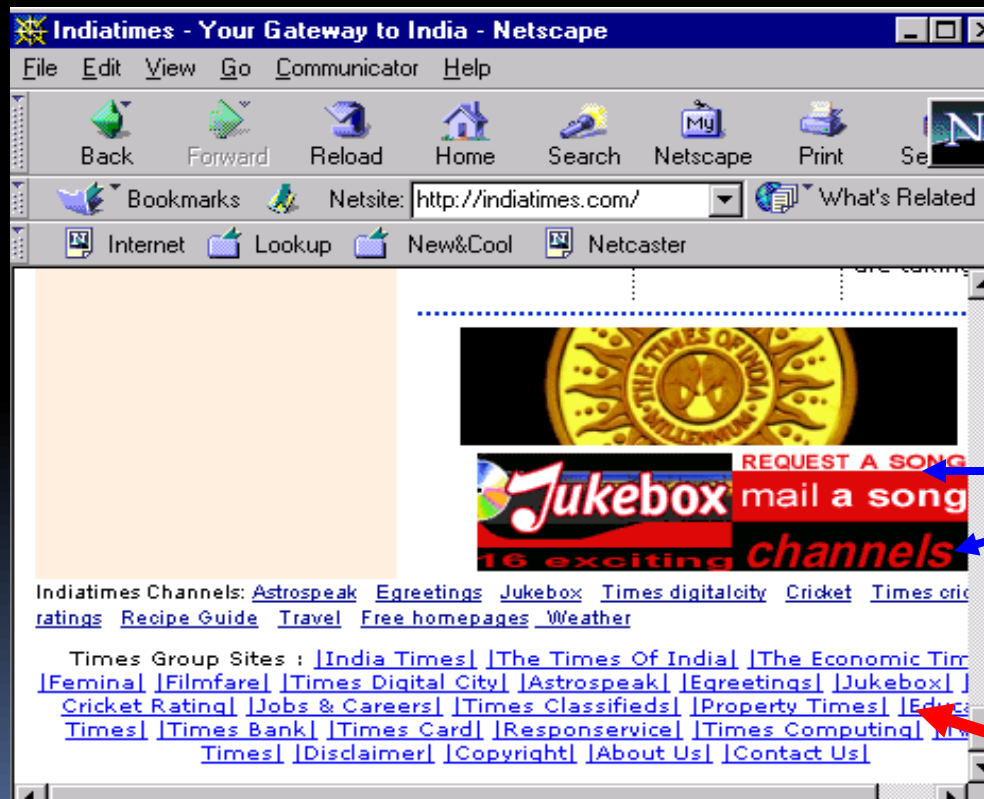
HTML Comments

- Comment tags `<!--` and `-->` are used to insert comments in HTML.
- Comments are not displayed by the browser, but they can help document your HTML.
- With comments you can place notifications and reminders in your HTML.

```
<!-- This is a comment -->  
<p>This is a paragraph.</p>  
<!-- Remember to add more information here -->
```

HTML Links

- Links are found in nearly all Web pages.
- Links allow users to click their way from page to page.



Graphical
Hyperlinks

Text Hyperlinks

HTML Hyperlinks (Links)

- The HTML `<a>` tag defines a hyperlink.
- A hyperlink (or link) is a word, group of words, or image that you can click on to jump to another document.
- When you move the cursor over a link in a Web page, the arrow will turn into a little hand.
- The most important attribute of the `<a>` element is the `href` attribute, which indicates the link's destination.

HTML Link Syntax

- The href attribute specifies the destination of a link.

`<a href="url">Link text</a>`

`<a href="http://www.newhorizons.com">New Horizons</a>`

- An unvisited link is underlined and blue
- A visited link is underlined and purple
- An active link is underlined and red

HTML Links - The id Attribute

- The id attribute can be used to create a bookmark inside an HTML document.
- Bookmarks are not displayed in any special way. They are invisible to the reader.

An anchor with an id inside an HTML document:

```
<a id="tips">Useful Tips Section</a>
```

Create a link to the "Useful Tips Section" inside the same or different document.

```
<a href="#tips">Visit the Useful Tips Section</a>
```

```
<a href="http://www.htmltips.com/html_links.htm#tips">
```

```
Visit the Useful Tips Section</a>
```



The HTML `<head>` Element

- The `<head>` element is a container for all the head elements.
 - Elements inside `<head>` can include scripts, instruct the browser where to find style sheets, provide meta information, and more.
 - The following tags can be added to the head section: `<title>`, `<style>`, `<meta>`, `<link>`, `<script>`, `<noscript>`, and `<base>`.
- 

The HTML `<meta>` Element

- Metadata is data (information) about data.
- The `<meta>` tag provides metadata about the HTML document.
- Metadata will not be displayed on the page, but will be machine parsable.
- Meta elements are typically used to specify page description, keywords, author of the document, last modified, and other metadata.
- The metadata can be used by browsers (how to display content or reload page), search engines (keywords), or other web services.
- `<meta>` tags always go inside the `<head>` element.

<meta> Tags - Examples

- Define keywords for search engines:

```
<meta name="keywords" content="HTML, CSS,  
XML, XHTML, JavaScript">
```

- Define a description of your web page:

```
<meta name="description" content="Free Web  
tutorials on HTML and CSS">
```

- Define the author of a page:

```
<meta name="author" content="Hege Refsnes">
```

- Refresh document every 30 seconds:

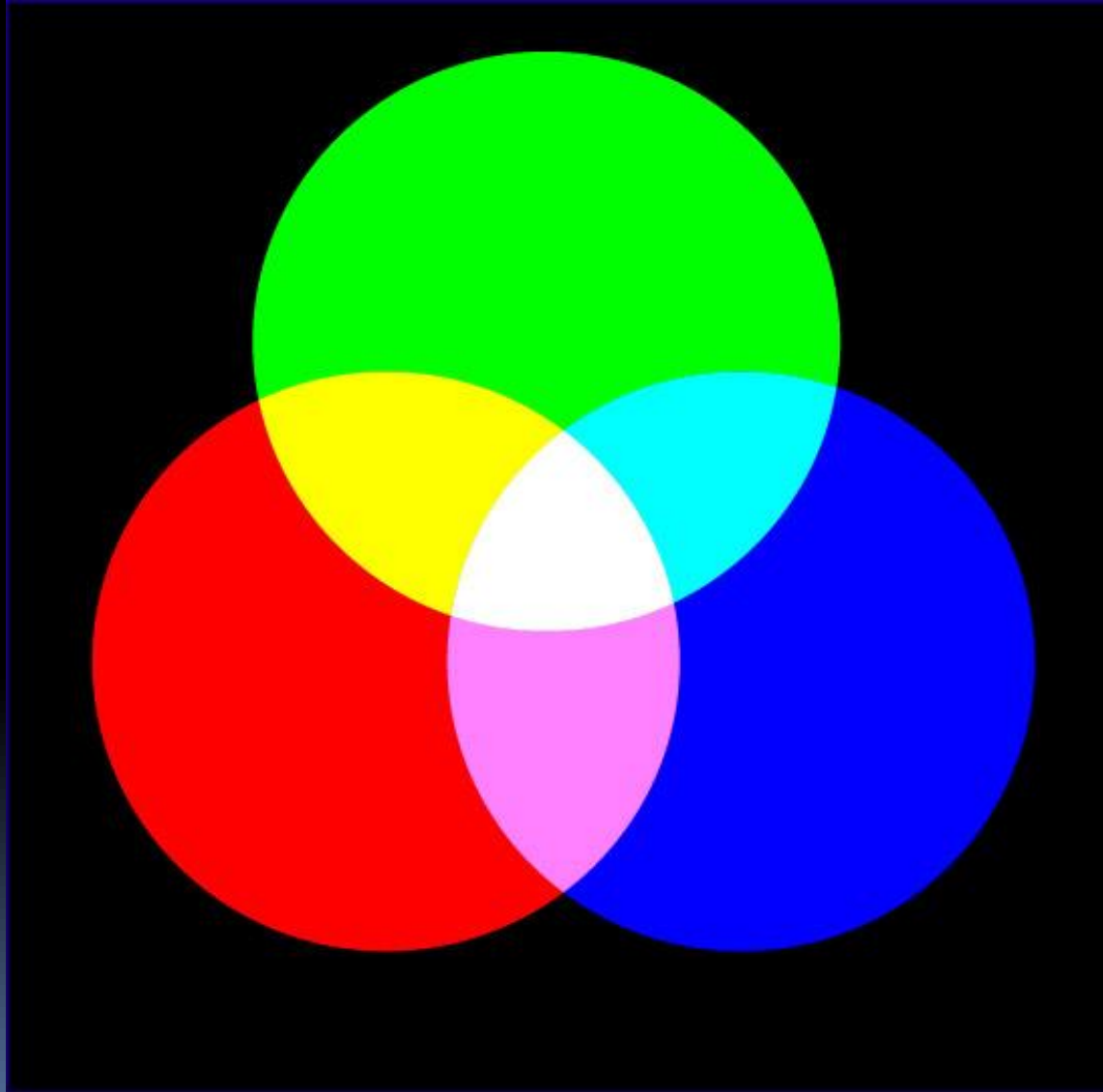
```
<meta http-equiv="refresh" content="30">
```



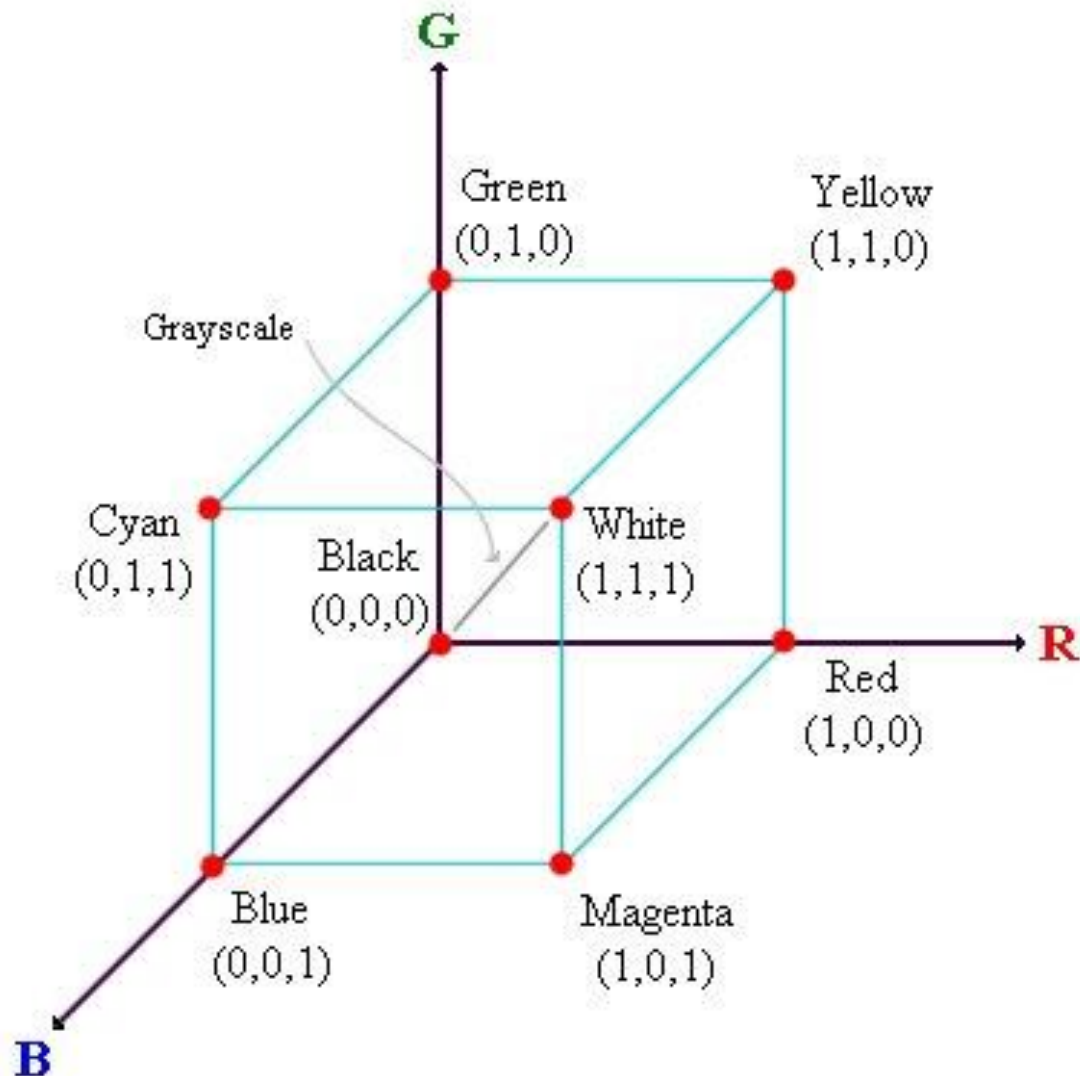

Color Code

- Colors are set using “**RGB**” color codes, which are, represented as hexadecimal values. Each 2-digit section of the code represents the amount, in sequence, of **red**, **green** or **blue** that forms the color. For example, a **RGB** value with 00 as the first two digits has no red in the color.

Main Colours



RGB Colour Model



16 Basic Colors

Color Name	RGB Triplet	Hexadecimal	Color Name	RGB Triplet	Hexadecimal
Aqua	(0,255,255)	00FFFF	Navy	(0,0,128)	000080
Black	(0,0,0)	000000	Olive	(128,128,0)	808000
Blue	(0,0,255)	0000FF	Purple	(128,0,128)	800080
Fuchsia	(255,0,255)	FF00FF	Red	(255,0,0)	FF0000
Gray	(128,128,128)	808080	Silver	(192,192,192)	C0C0C0
Green	(0,128,0)	008000	Teal	(0,128,128)	008080
Lime	(0,255,0)	00FF00	White	(255,255,255)	FFFFFF
Maroon	(128,0,0)	800000	Yellow	(255,255,0)	FFFF00

Color Codes

1. WHITE
2. BLACK
3. RED
4. GREEN
5. BLUE
6. MAGENTA
7. CYAN
8. YELLOW
9. AQUAMARINE
10. BAKER'S CHOCOLATE
11. VIOLET
12. BRASS
13. COPPER
14. PINK
15. ORANGE

1. #FFFFFF
2. #000000
3. #FF0000
4. #00FF00
5. #0000FF
6. #FF00FF
7. #00FFFF
8. #FFFF00
9. #70DB93
10. #5C3317
11. #9F5F9F
12. #B5A642
13. #B87333
14. #FF6EC7
15. #FF7F00

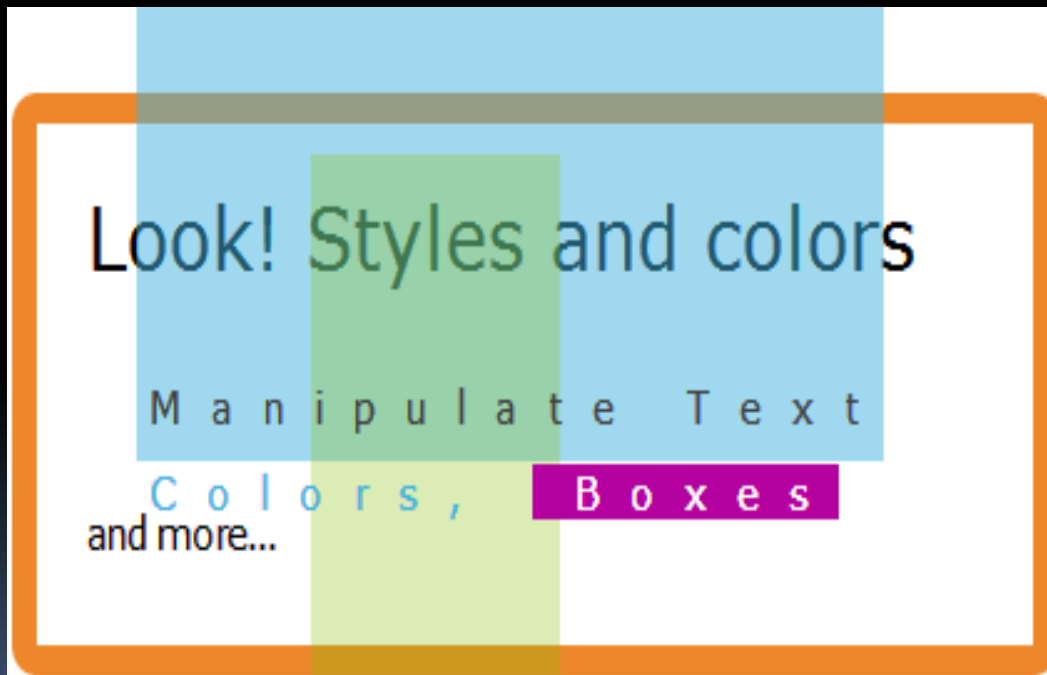
Color Codes

- If you require more information about color values, there is an excellent site entitled “VGDesign’s Interactive Color Cube” that displays the background color code when you put your cursor over a small color sample. The Web address is :

<http://www.vgdesign.com/color.html>

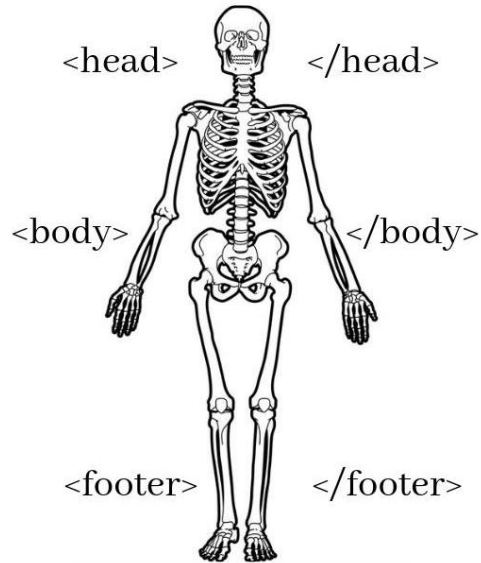
HTML Styles - CSS

- CSS (Cascading Style Sheets) is used to style HTML elements.



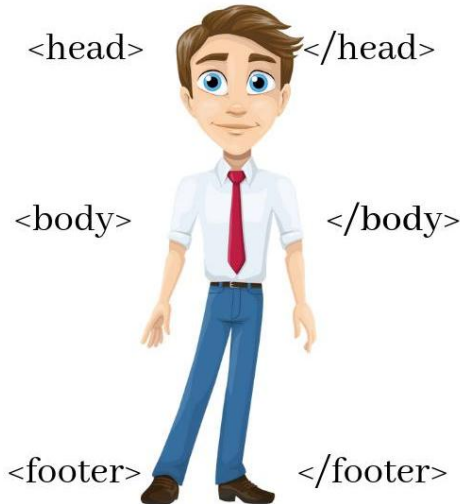
HTML Vs CSS

HTML

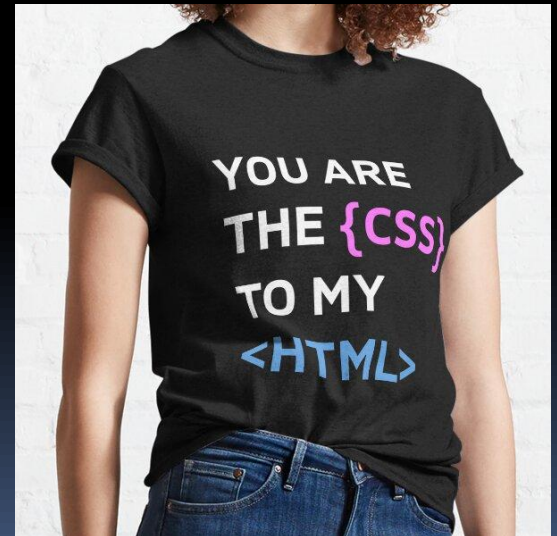


Structural Layer

HTML + CSS



Presentational Layer



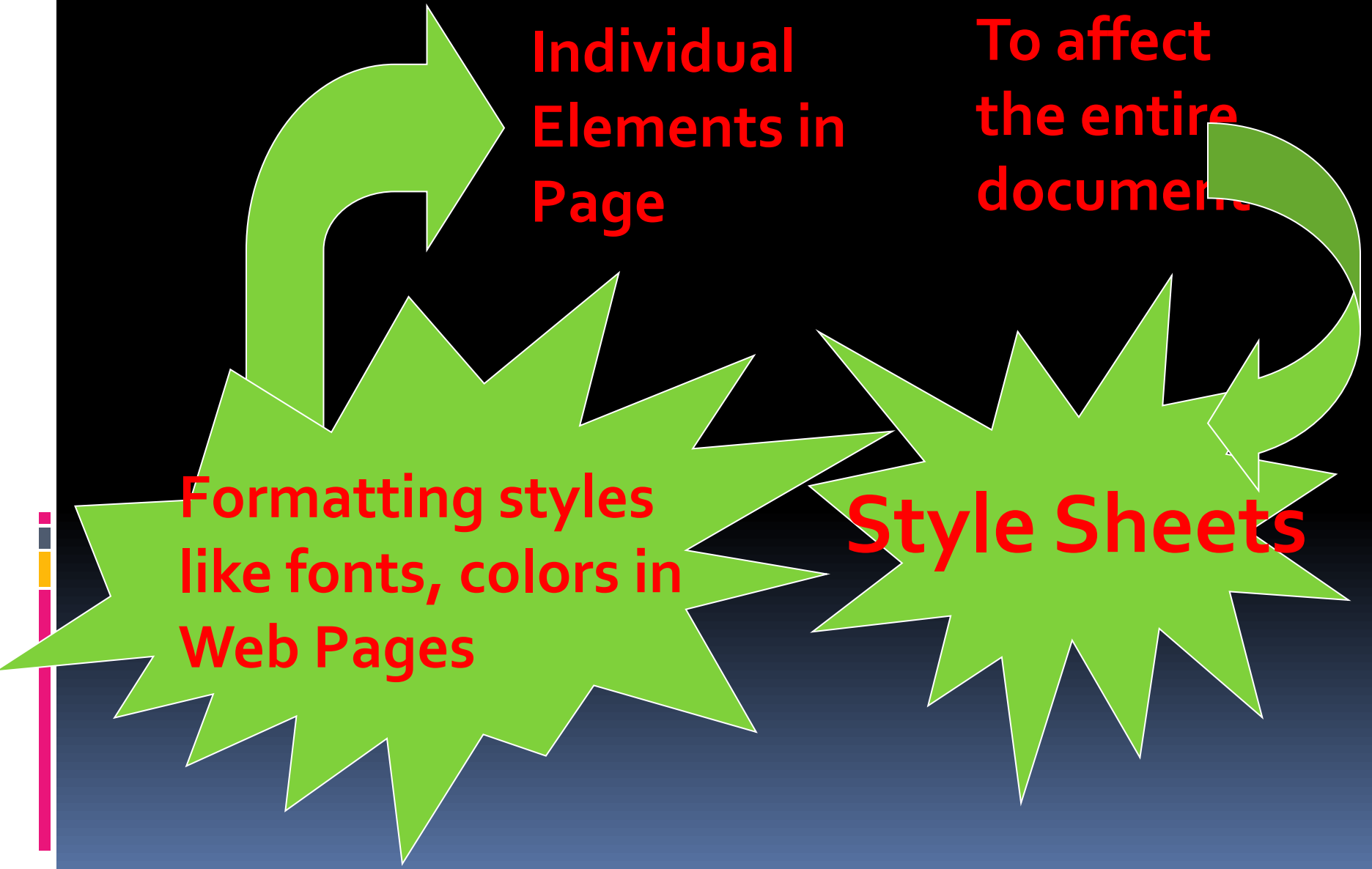
Style Sheets Introduction

Individual
Elements in
Page

To affect
the entire
document

Formatting styles
like fonts, colors in
Web Pages

Style Sheets



Style Sheet

Style Rules



Style Sheet - A
collection of
formatting styles

Selector Declaration

Syntax:

Selector {Declaration}

Property:Value



Example :

H1{Color:Blue}

Styling HTML with CSS

- CSS was introduced together with HTML 4, to provide a better way to style HTML elements.

CSS can be added to HTML in the following ways:

- Inline - using the style **attribute** in HTML elements
- Internal - using the `<style>` **element** in the `<head>` section
- External - using an external CSS **file**



Inline Styles

- An inline style can be used if a unique style is to be applied to one single occurrence of an element.
- To use inline styles, use the style attribute in the relevant tag. The style attribute can contain any CSS property.

```
<body style="background-color:yellow;">
<h2 style="background-color:red;">This is a heading</h2>
<p style="background-color:green;">This is a paragraph.</p>
<p style="font-family:arial;color:red;font-size:20px;">A
paragraph.</p>
<h1 style="text-align:center;">Center-aligned heading</h1>
```

- **The background-color property defines the background color for an element**

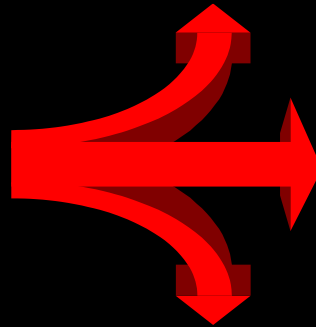
Internal Style Sheet

- An internal style sheet can be used if one single document has a unique style.
- Internal styles are defined in the <head> section of an HTML page, by using the <style> tag

```
<head>  
<style>  
body {background-color:yellow;}  
p {color:blue;}  
</style>  
</head>
```

Types of Selectors

Selectors



Simple Selector

HTML Selector

Class Selector

ID Selector

Simple Selector

<P> {Color:Blue}

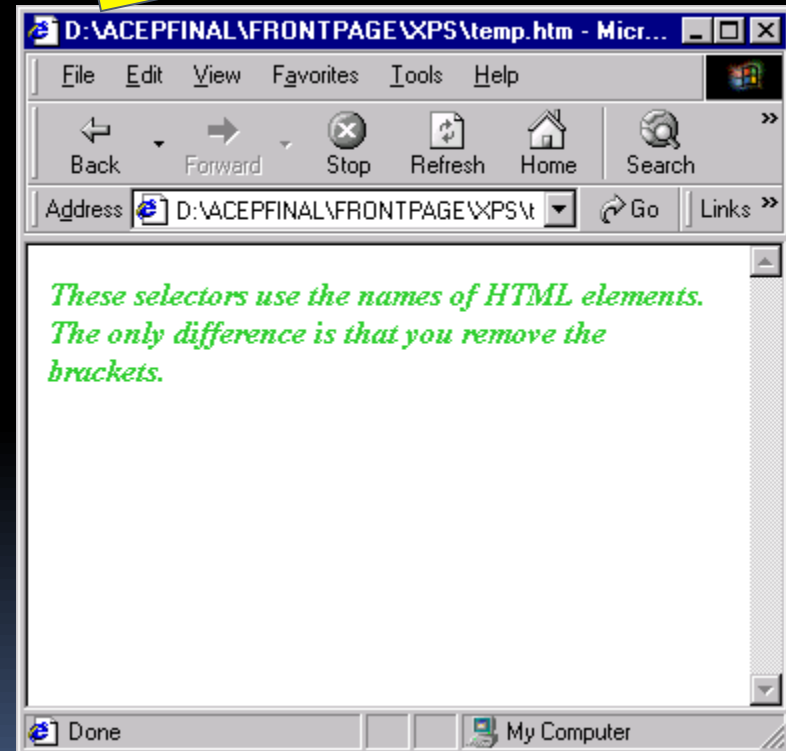
HTML Selector

P {Color:Blue}

Uses names of HTML Elements

HTML Selector

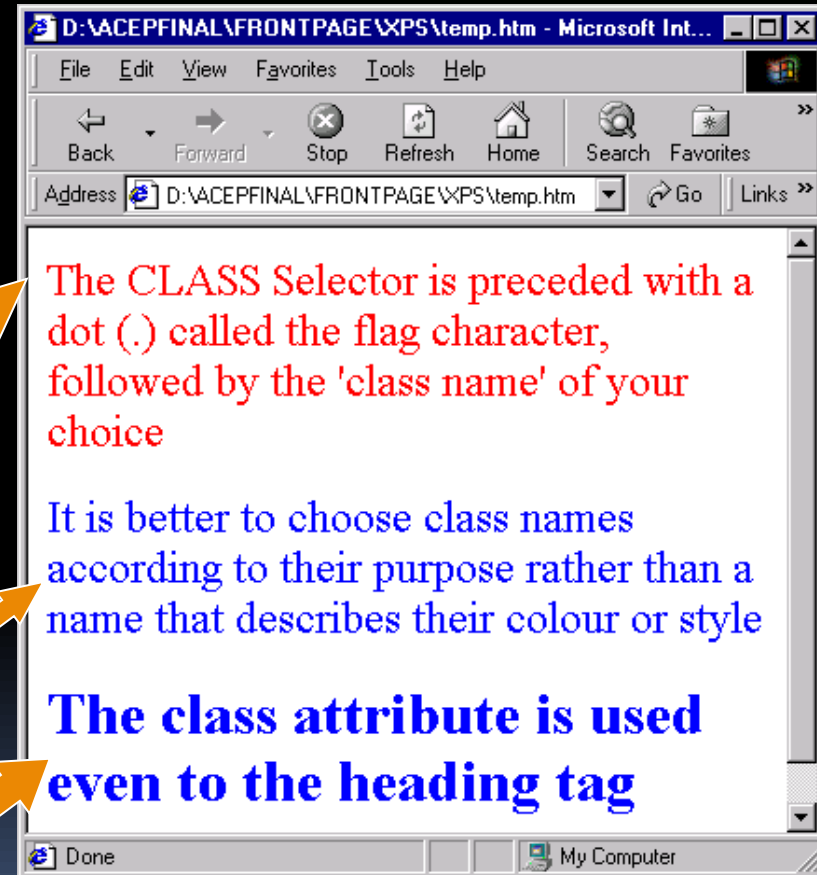
```
<HTML>
<HEAD>
<STYLE TYPE="text/css">
  P {font-style:italic;
    fontweight:bold;
    color:limegreen}
</STYLE>
</HEAD>
<BODY>
  <P>These selectors use the names of
    HTML elements. The only
    difference is that you remove the
    brackets.
  </P>
</BODY>
</HTML>
```



Class Selector

- To apply styles to specific parts of a document

```
<HEAD>
<STYLE TYPE="text/css">
    .note { color: blue }
    .syntax { color: red }
    p.syntax {color:red}
    P{font-size: large}
</STYLE>
</HEAD>
<BODY>
<P class="syntax">The CLASS Selector is
    preceded with a dot (.) called the flag
    character, followed by the 'class name'
    of your choice
<P class="note">It is better to choose class names
    according to their purpose rather than
    a name that describes their colour or
    style
<H1 class="note"> The class attribute is used even
    to the heading tag
<H1>
</HTML>
```



ID Selectors

- An ID selector is preceded by a hash (#) mark and to apply ID selector the ID attribute of an HTML element is used

```
<HEAD>
<STYLE TYPE="text/css">
    #control { color: red }
</STYLE>

<H2 id="control">Fire is this
    colour</H2>
<H2>The browser controlled display
    of an H2 heading</H2>
<P id="control">Applying an ID to a
    paragraph element</P>
<P>This paragraph has no style
    applied</P>
</BODY>
</HTML>
```



External Style Sheet

- An external style sheet is ideal when the style is applied to many pages.
- With an external style sheet, you can change the look of an entire Web site by changing one file.
- Each page must link to the style sheet using the <link> tag.
- The <link> tag goes inside the <head> section

```
<head>
```

```
<link rel="stylesheet" type="text/css" href="mystyle.css">
```

```
</head>
```

Cascading Style Sheets

- Style rules can be saved in a single file called style sheet to apply common style to many documents
- Style sheet can then be imported or linked.

REL=STYLE SHEET -
Relation to the
external document in a
style sheet.

<HTML>

<HEAD>

<LINK REL=STYLE SHEET TYPE="TEXT/CSS"
HREF="PATH WHERE THE STYLE SHEET IS SAVED"

</HEAD>

BODY TAGS

</HTML>

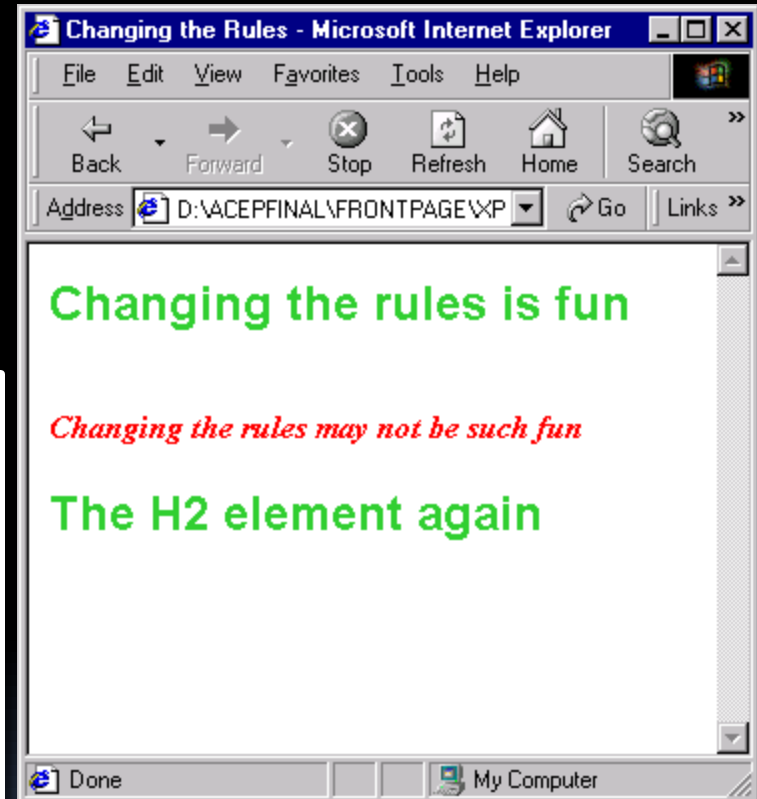
- The method of linking or importing a Style Sheet is called as Cascading Style Sheet

Linking Style Sheets

Style1.CSS

```
H2 {color:limegreen;
    font-family:"Arial";
    font-size:normal;
}
#warning {color:red;
          font-weight:bold;
          font-style:italic;
}
```

```
<!DOCTYPE HTML>
<HEAD>
<TITLE> Changing the Rules</TITLE>
<LINK REL=stylesheet HREF="style1.css"
      TYPE="text/css">
</HEAD>
<H2>Changing the rules is fun</H2>
<BR>
<P id="warning">Changing the rules may
  not be such fun
<H2>The H2 element again<P>
</HTML>
```



HTML Images

Norwegian Mountain Trip



The Tag and the Src Attribute

- In HTML, images are defined with the tag.
- The tag is empty, which means that it contains attributes only, and has no closing tag.
- To display an image on a page, you need to use the src attribute.
- Src stands for "source".
- The value of the src attribute is the URL of the image you want to display.

Syntax for defining an image

```

```

- The URL points to the location where the image is stored.
- The required alt attribute specifies an alternate text for an image, if the image cannot be displayed

Set Height and Width of an Image

- The height and width attributes are used to specify the height and width of an image.
- The attribute values are specified in pixels by default

```

```




HTML Blocks

- HTML elements can be grouped together with `<div>` and `<span>`.
- Most HTML elements are defined as **block level** elements or as **inline** elements.
- Block level elements normally start (and end) with a new line when displayed in a browser.
- Examples: `<h1>`, `<p>`, `<ul>`, `<table>`

HTML Inline Elements

- Inline elements are normally displayed without starting a new line.
- Examples: `<b>`, `<td>`, `<a>`, `<img>`

The HTML <div> Element

- The HTML <div> element is a block level element that can be used as a container for grouping other HTML elements.
- It is a block level element, the browser will display a line break before and after it.
- When used together with CSS, the <div> element can be used to set style attributes to large blocks of content.
- Defines a section in a document (block-level)



`<div>` Element

- Another common use of the `<div>` element, is for document layout.
 - It replaces the "old way" of defining layout using tables.
 - Using `<table>` elements for layout is not the correct use of `<table>`.
 - The purpose of the `<table>` element is to display tabular data.
- 



The HTML `<span>` Element

- Defines a section in a document (inline)
 - The HTML `<span>` element is an inline element that can be used as a container for text.
 - The `<span>` element has no special meaning.
 - When used together with CSS, the `<span>` element can be used to set style attributes to parts of the text.
- 



HTML Layouts

- Web page layout is very important to make your website look good.
 - Design your webpage layout very carefully.
 - Most websites have put their content in multiple columns (formatted like a magazine or newspaper).
 - Multiple columns are created by using `<div>` or `<table>` elements.
 - CSS are used to position elements, or to create backgrounds or colorful look for the pages.
- 

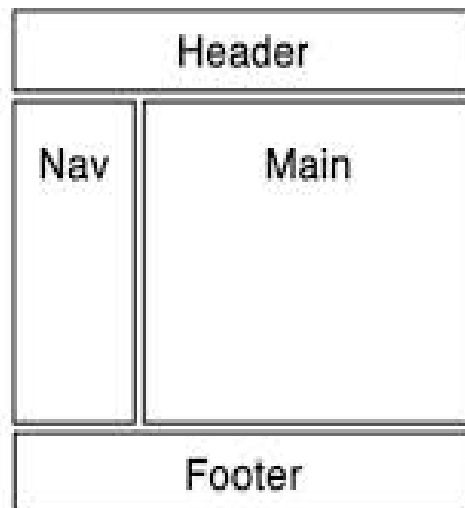


HTML Layouts - Using <div> Elements

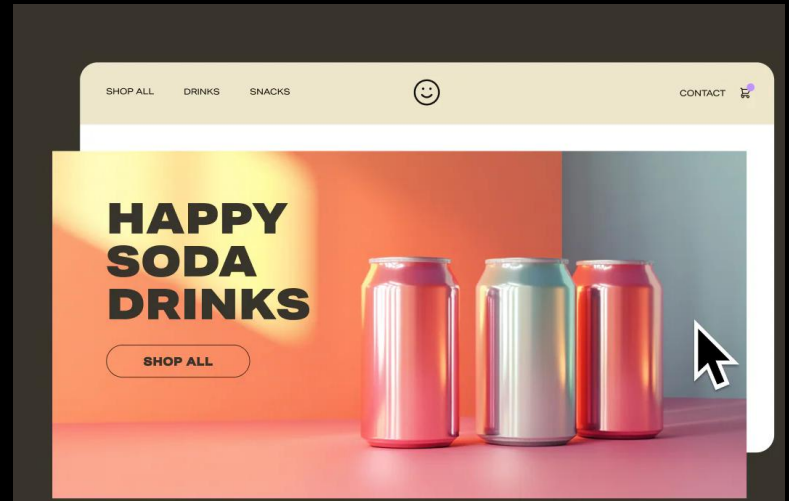
- The div element is a block level element used for grouping HTML elements.
 - The biggest advantage of using CSS is that, if you place the CSS code in an external style sheet, your site becomes MUCH EASIER to maintain.
 - You can change the layout of all your pages by editing one file.
- 

Layout - Templates

Basic HTML Layout of Web Page



Hero Image



- In HTML, a "hero section" refers to the prominent and visually striking area located at the top of a webpage.
- It is typically the first element a user sees upon landing on a site and serves as a crucial component for making a strong first impression and conveying the website's core message or purpose.

Purpose of a hero section

Captivate Attention: Immediately engage visitors and make a positive first impression.

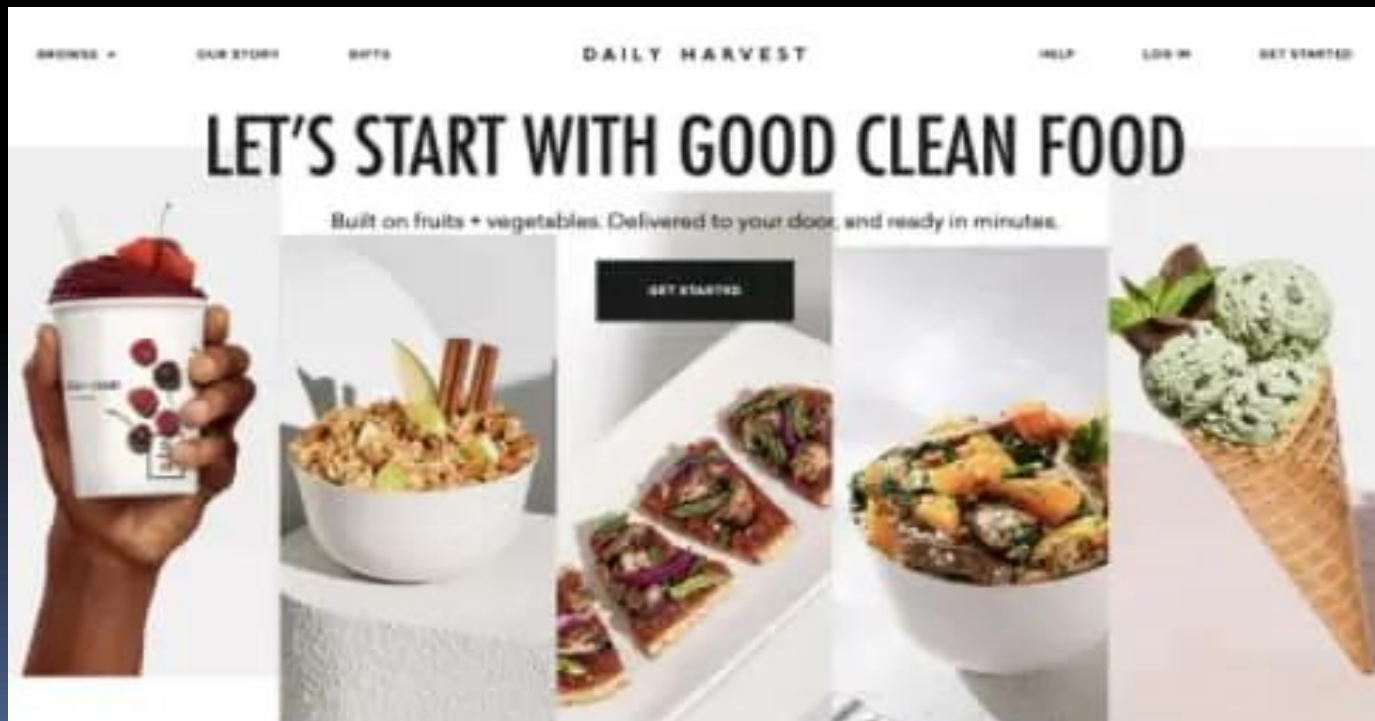
Convey Value: Quickly communicate the essence of the website's content or offerings.

Guide Users: Direct visitors towards key information or desired actions.

Establish Brand Identity: Reinforce the brand's visual style and message.

Hero Section

- A Hero Image is a large image with text, often placed at the top of a webpage



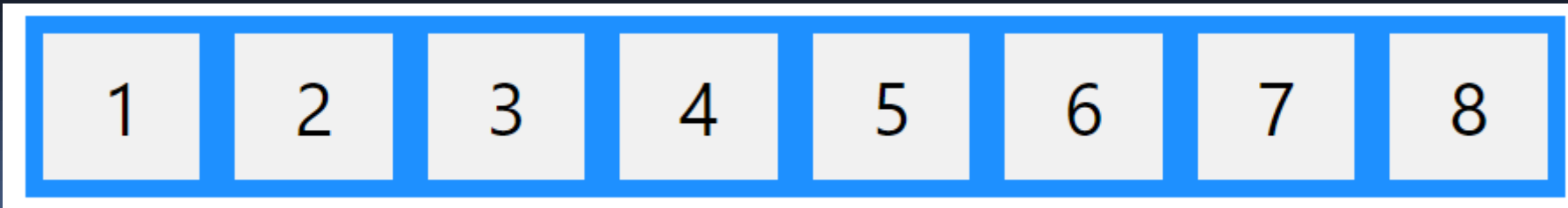
Demo

```
<div class="hero-image">
  <div class="hero-text">
    <h1>I am John Doe</h1>
    <p>And I'm a Photographer</p>
    <button>Hire me</button>
  </div>
</div>
```

After that Apply CSS

Flexbox

- Flexbox is a one-dimensional layout method for arranging items in rows or columns.
- Items **flex (expand)** to fill additional space or shrink to fit into smaller spaces
- To use flexbox, you apply **display: flex** to the parent element of the elements you want to lay out;
- All its direct children then become flex items.



CSS Flexbox Layout Module

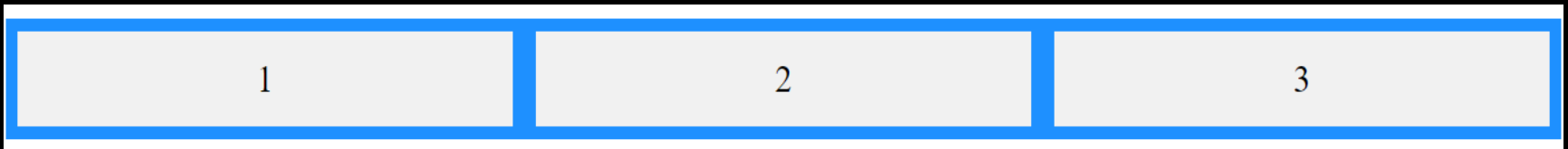
Before the Flexbox Layout module, there were four layout modes:

- Block, for sections in a webpage
- Inline, for text
- Table, for two-dimensional table data
- Positioned, for explicit position of an element

The Flexible Box Layout Module, makes it easier to design flexible responsive layout structure without using float or positioning.

Flexbox Elements

- To start using the Flexbox model, you need to first define a **flex container**.
- The element below represents a flex container (the blue area) with three flex items.



```
<div class="flex-container">
  <div>1</div>
  <div>2</div>
  <div>3</div>
</div>
```

```
.flex-container {
  display: flex;
  background-color: DodgerBlue;
}
.flex-container > div { flex: 1;
  background-color: #f1f1f1;
  margin: 10px; text-align:center
  padding: 20px;
  font-size: 30px; }
```

Understand '+', '>' symbols in CSS Selector

- **> Sign:**
- It will target elements which are DIRECT children of a particular element.

```
div#container > p {  
  border: 1px solid black; }
```

+ Sign:

- It is Adjacent sibling combinator. It combines two sequences of simple selectors having the same parent and the second one must come IMMEDIATELY after the first.

```
div + p {  
  color: green;  
}
```


Forms

A form is a collection of fields on a page.

The screenshot shows a web browser window titled "Register - Microsoft Internet Explorer". The address bar displays "C:\WINNT\Profiles\GUNEET\Desktop\Shoppingmall\register.htm". The page content features a red header with the text "Register with us" and a yellow pencil icon. Below the header, there is a navigation bar with "All Products" and a "GO!" button, and a link "click here to see your shopping cart ->". The main form area is titled "Apna Bazaar" and contains several sections: "Login" with a text field and an asterisk, "Password" with a text field and an asterisk, "Re-enter" with a text field and an asterisk, "Full Name" with a text field and an asterisk, "Sex" with radio buttons for "Male" and "Female", "Birth Date" with a date picker, and "Address" with a text field. A red circle highlights the form fields, and an arrow points from the text "Form Fields" to the circle.

Form Handler

- Form handler collects information from the form. It is used to:
 - **Store the data submitted by the user into a text file/database. The data can be retrieved as and when required.**
 - **Update a database with the data submitted by the user.**
 - **Post the data on a Web page**

Creating Forms

Forms are classified into three:

- Form Header
- Input Fields
- Action Buttons

<Form>....</Form> tag

Syntax:

<HTML>

<BODY>

<FORM>

Form

Elements

</FORM>

</BODY>

<HTML>

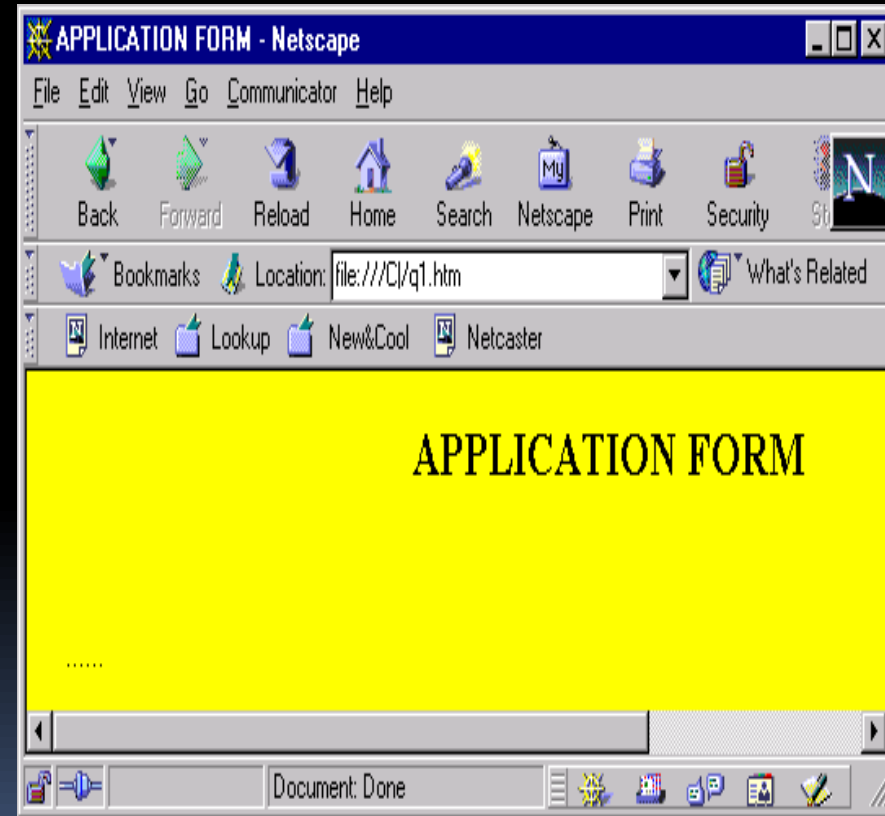
Creating Form Header

Example- creating a simple form header using <Caption> Tag



```
<HTML>
<HEAD>
  <TITLE> Application Form </TITLE>
</HEAD>
<BODY bgcolor="gray" >
  <TABLE CellPadding=10 Width="700">
<CENTER>
<CAPTION ALIGN="TOP">
  <H2>APPLICATION FORM</H2>
</CAPTION>
</CENTER>
</TABLE>
</HTML>
```

Form Elements are
added using Table



The `<label>` Element

- The `<label>` tag defines a label for many form elements.
- The `<label>` element is useful for screen-reader users, because the screen-reader will read out loud the label when the user focus on the input element.
- The `<label>` element also help users who have difficulty clicking on very small regions (such as radio buttons or checkboxes) - because when the user clicks the text within the `<label>` element, it toggles the radio button/checkbox.
- The `for` attribute of the `<label>` tag should be equal to the `id` attribute of the `<input>` element to bind them together.

The <label> Element

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname"><br>
  <label for="lname">Last name:</label><br>
  <input type="text" id="lname" name="lname">
</form>
```

Try it Yourself »

This is how the HTML code above will be displayed in a browser:

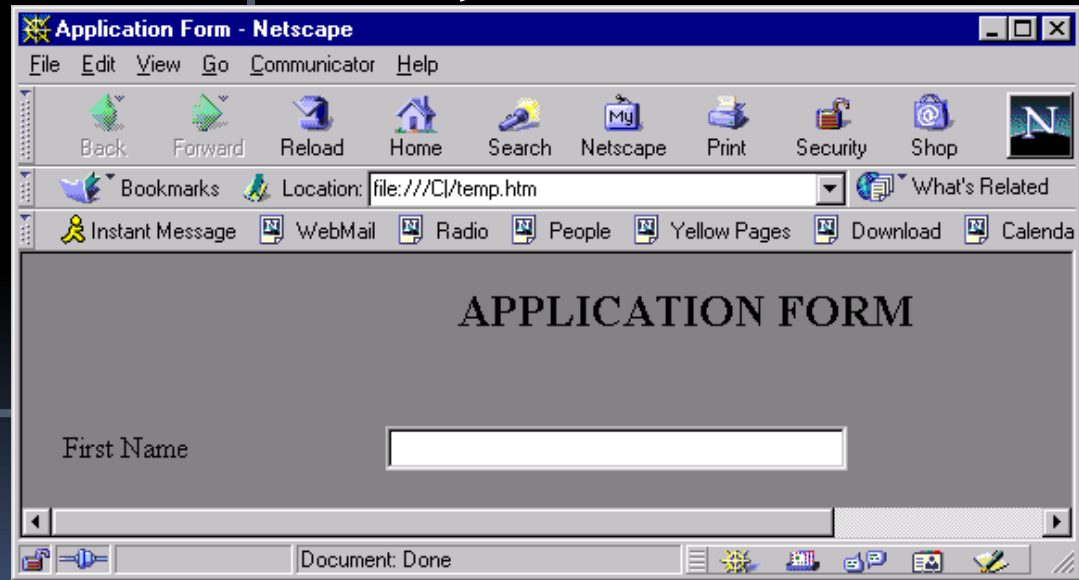
First name:

Last name:

Text Input Type

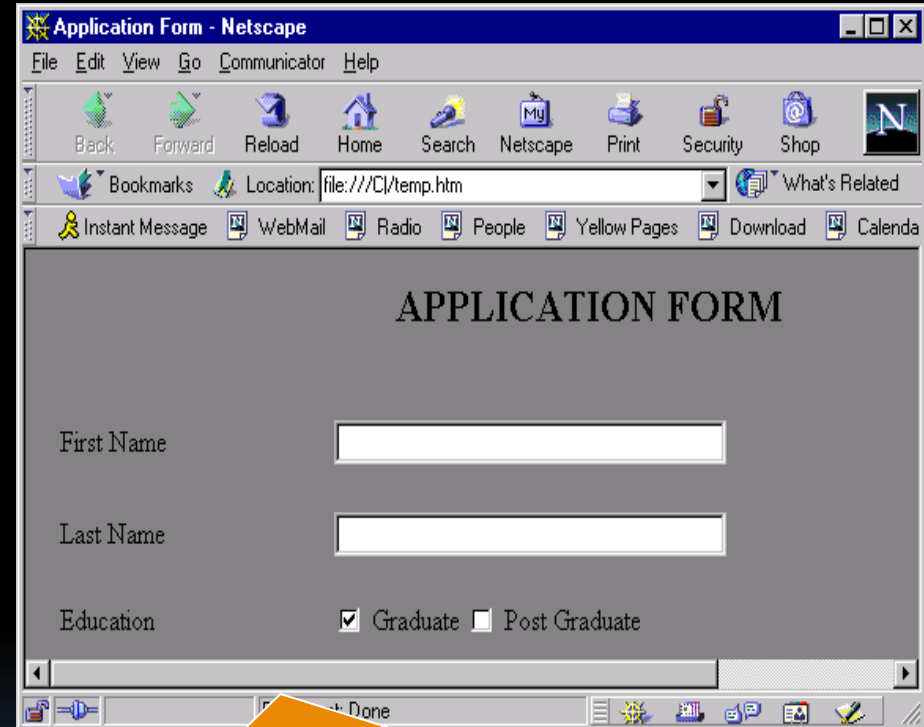
Example

```
<FORM>
<FONT FACE="COURIER NEW">
<FONT SIZE="4">
<TR>
<TD>First Name</TD>
<TD><INPUT TYPE = "TEXT"
      NAME = "EName1"
      SIZE = "30"
      MAXLENGTH = "30"
      VALUE = "">
</TD>
</TR>
```



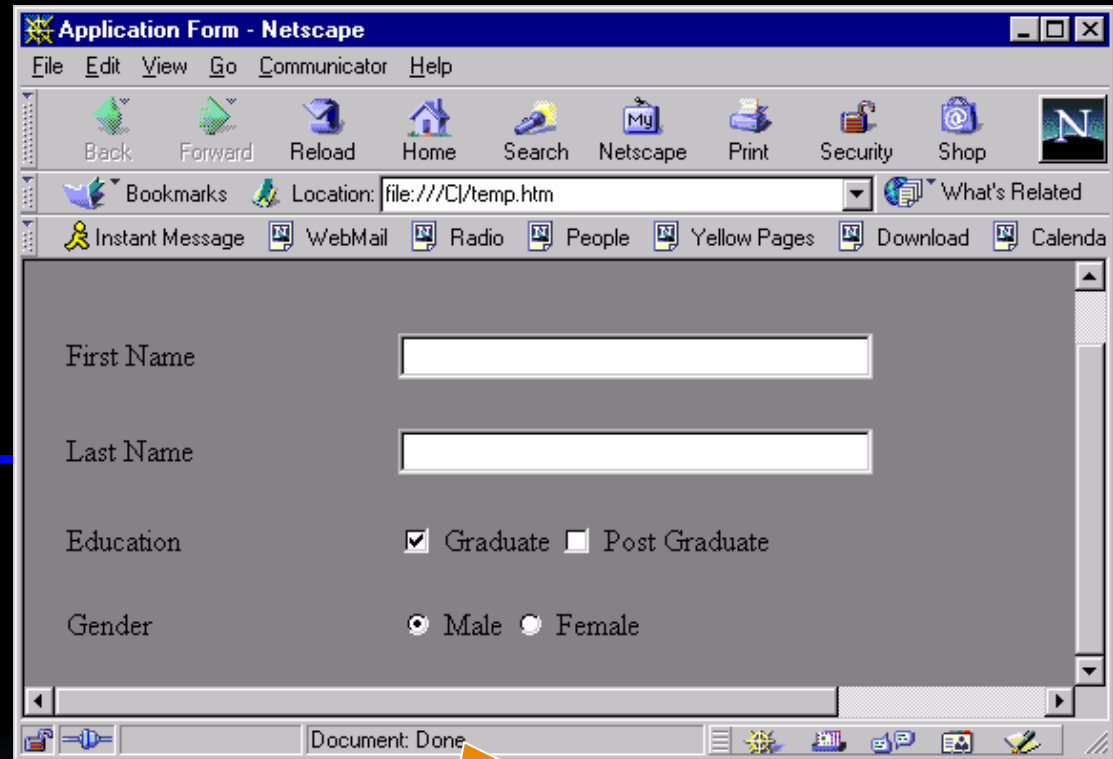
Checkbox Input Type

```
<TR>
<TD>Education</TD>
<TD>
<INPUT TYPE = "CHECKBOX"
      NAME = "Education"
      VALUE = "Graduate" CHECKED>
      Graduate
<INPUT TYPE = "CHECKBOX"
      NAME = "Education"
      VALUE = "Post Graduate">
      Post Graduate
</TD>
</TR>
```



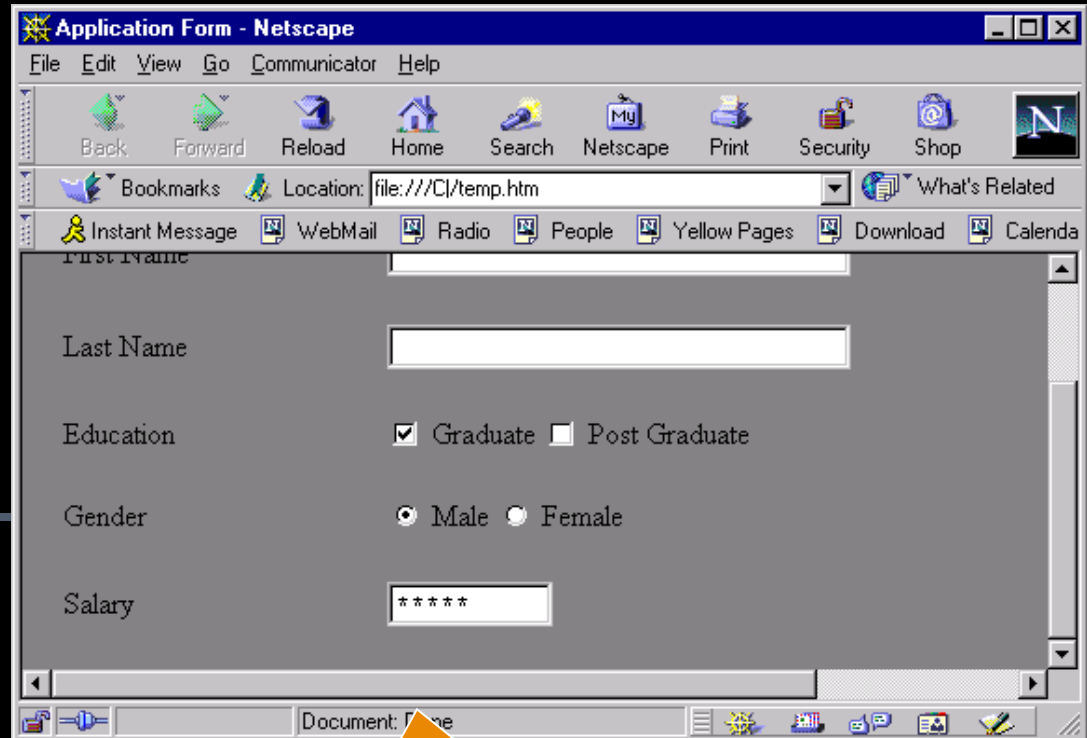
Radio Button Input Type

```
<TR>
<TD>Gender</TD>
<TD>
<INPUT TYPE = "RADIO"
  NAME = "Gender"
  VALUE = "Male" CHECKED> Male
<INPUT TYPE = "RADIO"
  NAME = "Gender"
  VALUE = "Female" > Female
</TD>
</TR>
```



Password Input Type

```
<TR>  
<TD>Salary</TD>  
<TD>  
<INPUT TYPE = PASSWORD"  
  NAME= "Salary"  
  SIZE = "10">  
</TD>  
</TR>
```



The screenshot shows a Netscape browser window titled "Application Form - Netscape". The address bar displays "file:///C:/temp.htm". The form contains the following fields:

- First Name:
- Last Name:
- Education: ☒ Graduate ☐ Post Graduate
- Gender: ☒ Male ☐ Female
- Salary:

A large orange arrow points from the "Salary" field in the form to the corresponding HTML code in the left panel.

Multiple Line Text Input

- **<TEXTAREA>.....**
<TEXTAREA> Tag
- **TEXTAREA Tag Attributes**
 - **Rows**
 - **Cols**
 - **Wrap**
 - **OFF**
 - **VIRTUAL**
 - **PHYSICAL**

Syntax

```
<TEXTAREA ROWS=" "
COLS=" " WRAP=" " >
```

Pull Down Menus

- `<SELECT>.....</SELECT>` Tag
 - `<OPTION>.....</OPTION>` Tag
- Syntax

```
<SELECT NAME="ListName">
```

```
<OPTION>Option1
```

```
<OPTION>Option2
```

```
</SELECT>
```

Action Buttons

- Submit Button

- Attributes

- TYPE

- VALUE

- Reset Button

- Attributes

- TYPE

- VALUE

Syntax

```
<TR>
```

```
<INPUT TYPE = "SUBMIT"  
VALUE = "SUBMIT">
```

```
<INPUT TYPE = "RESET"  
VALUE = "RESET">
```

```
</TD>
```

```
</TR>
```

The Action Attribute

- The action attribute defines the action to be performed when the form is submitted.
- Normally, the form data is sent to a web page on the server when the user clicks on the submit button.
- If the action attribute is omitted, the action is set to the current page.

```
<form action="/action_page.jsp">
```

The Method Attribute

- The method attribute specifies the HTTP method (GET or POST) to be used when submitting the form data

```
<form action="/action_page.jsp" method="get">
```

```
<form action="/action_page.jsp" method="post">
```

When to Use GET?

- The default method when submitting form data is GET.
- However, when GET is used, the submitted form data will be visible in the page address field
- Appends form-data into the URL in name/value pairs
- The length of a URL is limited (about 3000 characters)
- Never use GET to send sensitive data! (will be visible in the URL)
- Useful for form submissions where a user wants to bookmark the result
- GET is better for non-secure data, like query strings in Google

`action_page.jsp?firstname=Mickey&lastname=Mouse`



When to Use POST?

- Always use POST if the form data contains sensitive or personal information.
 - The POST method does not display the submitted form data in the page address field.
 - POST has no size limitations, and can be used to send large amounts of data.
 - Form submissions with POST cannot be bookmarked
- 

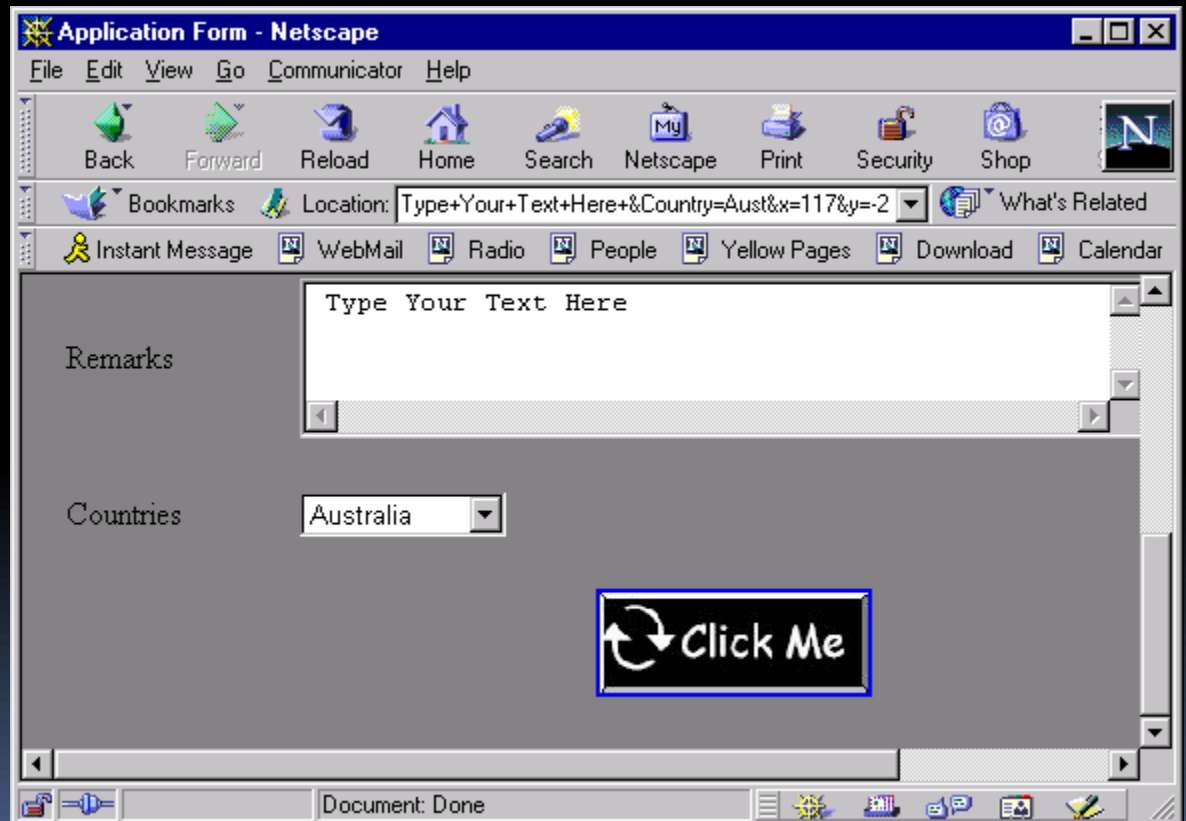
Image Input Type

Syntax

`<INPUT TYPE = "IMAGE" SRC = "Image Name">`

Example

`<INPUT TYPE = "IMAGE"
SRC = "ClickButton.jpg">`



FORM Tag

- Form Tag
 - Method Attribute
 - Action Attribute
- Syntax

<FORM

ACTION="MAILTO:*emailID*"

METHOD="POST" ***NAME***="NAME OF THE
FORM">

..... . .

</FORM>

WEB Databases

■ Need For Database Applications

- u Keep track of the customers who visit the site frequently to provide effective promotional schemes like discounts, Reward points based on the purchase amount that can be refunded later.
- u Provide customized information to customers, suppliers and employees
- u To hold on line quizzes, puzzles to encourage customer participation.
- u Keep track of the products sold
- u To maintain on-line catalogs